

ENGINEERING PROJECT PORTFOLIO

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Mechanical Engineering · Lancaster University · Expecting a First

A selection of design, simulation and build projects spanning automotive and high-performance engineering.

Adaptive Electromagnetic Valve Timing (AEVT)

01

MEng Individual Project · Oct 2025 – May 2026

OVERVIEW

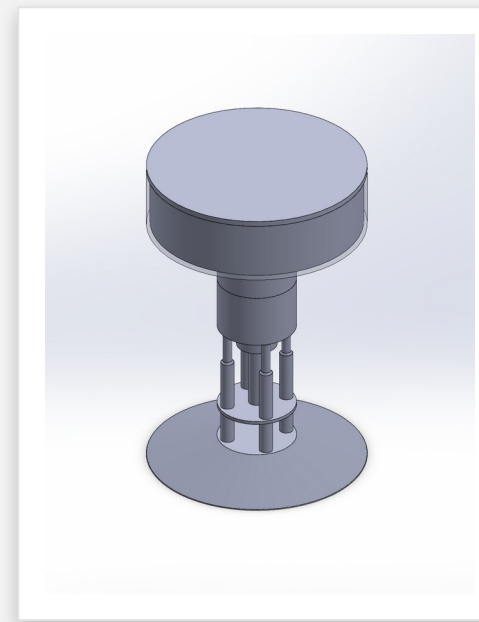
A solenoid-based, cam-less valve actuation prototype developed as a fully variable alternative to conventional camshaft-driven valve trains. A 12 V solenoid drives the valve to produce up to 15 mm of lift with spring return, while an Arduino control system using MOSFET switching and 4N35 optocoupler isolation generates variable-frequency PWM signals to simulate engine throttle and speed.

MY CONTRIBUTION

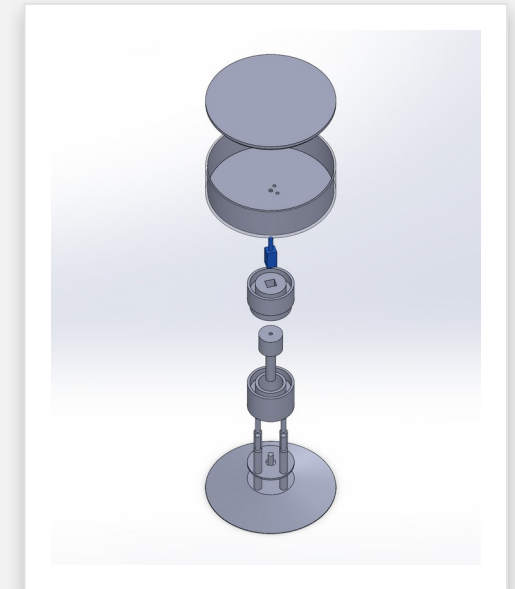
End-to-end mechanical design, material selection, circuit design and system testing, demonstrating the feasibility of electronically variable valve timing for improved efficiency and reduced pumping losses.



Build prototype



CAD assembly



Exploded view

Machine Learning-Based Adaptive Train Suspension

02

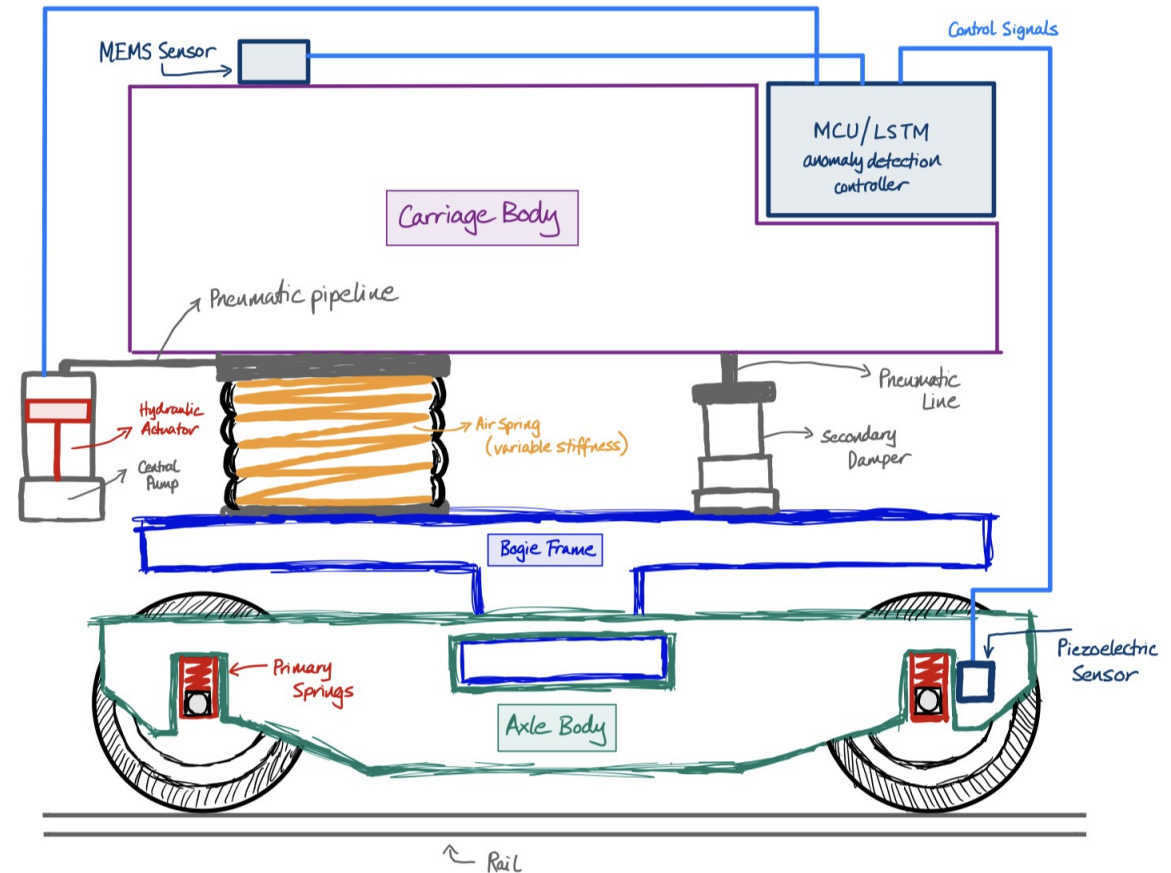
Group Project · Jan 2026 – Apr 2026

OVERVIEW

An ML-enhanced adaptive electronic suspension system for high-speed rail carriages. A two-degree-of-freedom mechanical model was formulated and an LSTM autoencoder trained on healthy vibration data to detect track anomalies in real time.

MY CONTRIBUTION

Designed a coordinated safety response combining rapid secondary damping adjustment with automatic speed reduction, shifting suspension behaviour from reactive control to predictive, data-driven fault detection.



System architecture and anomaly-detection sensing

Biofuel Storage - AC Adsorption Measurement System

03

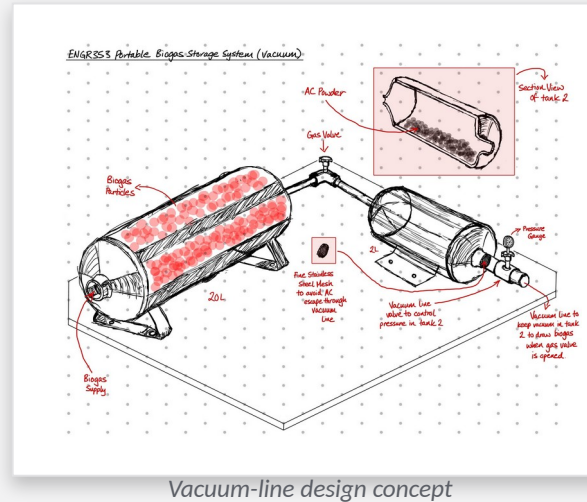
Group Design Project · Oct 2025 – Dec 2025

OVERVIEW

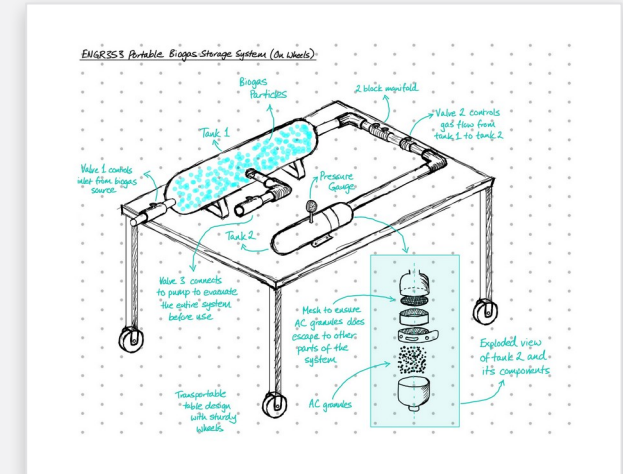
A portable, low-cost system for measuring the adsorption fill-up behaviour of activated carbon used in biogas transport. The dual-container pressure vessel arrangement enables methane storage testing at up to 12 bar within a £2,000 budget, sized to test 200 g of activated carbon and transportable over rough terrain by a single person.

MY CONTRIBUTION

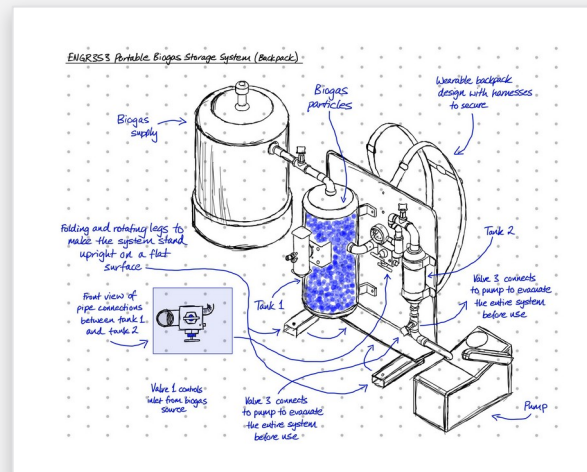
Produced the design sketches and led concept selection for the final arrangement.



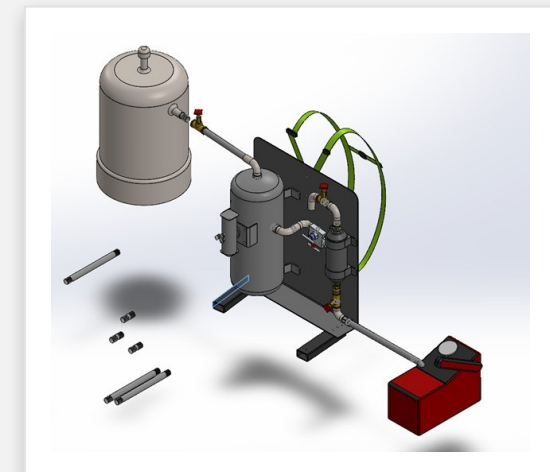
Vacuum-line design concept



Transportable wheeled design



Design iteration



Final CAD arrangement

Autonomous Track-Following Robot

Group Project · Oct 2024 – Mar 2025

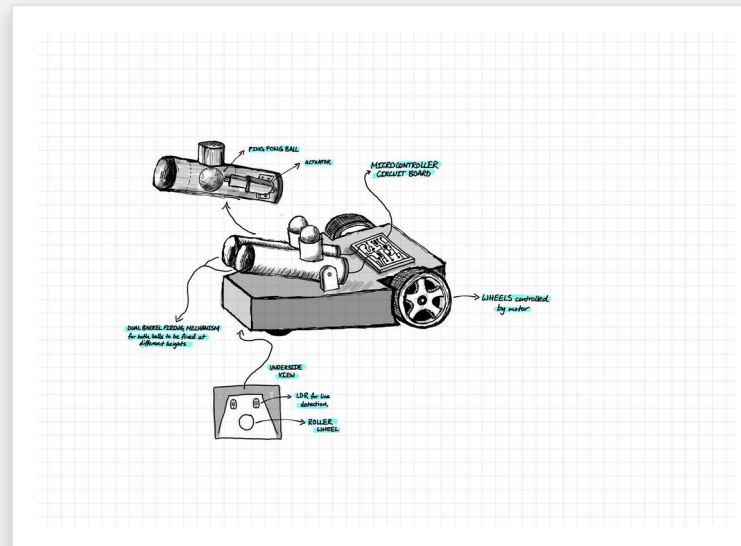
04

OVERVIEW

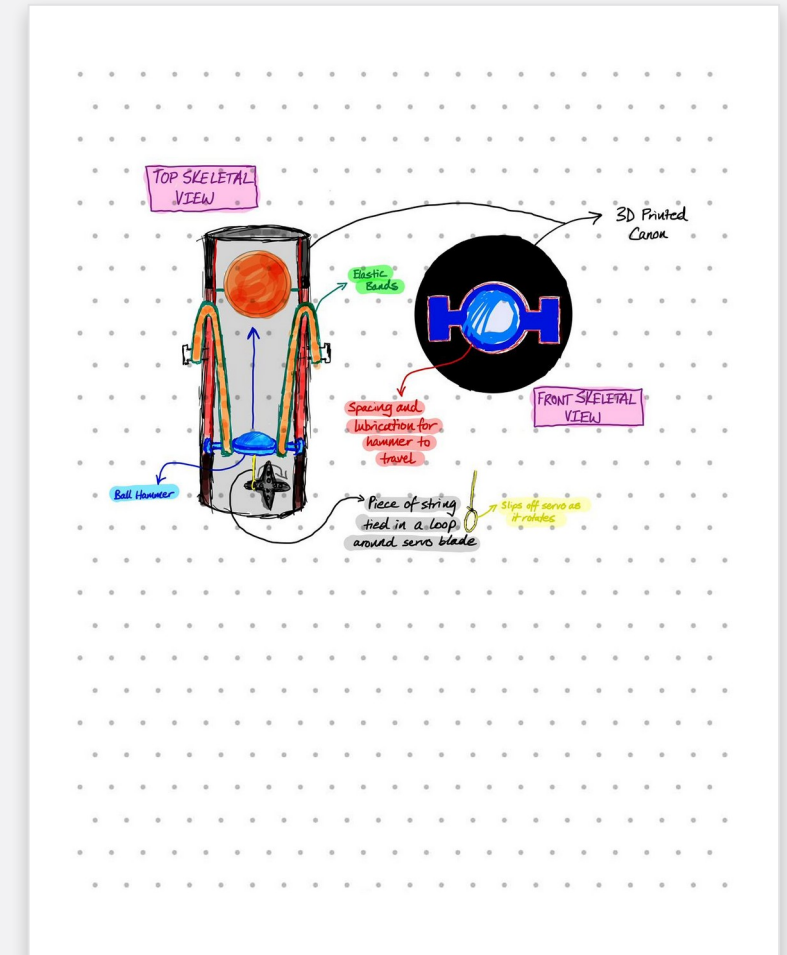
An autonomous robot designed, built and programmed to follow a 15 m track, pass under a bridge and accurately launch two ping-pong balls into target zones at different distances. Twin light-dependent resistors in a Wheatstone bridge feed an op-amp and H-bridge driven motors, refined with PID control for smooth line following, with a servo-actuated elastic band mechanism that fires on wall contact.

MY CONTRIBUTION

Contributed to CAD modelling, PCB soldering and structured testing throughout development.



Chassis and sensor layout



Ball-launcher mechanism

IMechE Foundation Design Challenge

Autonomous Vehicle · Group Project · Oct 2023 – Dec 2023

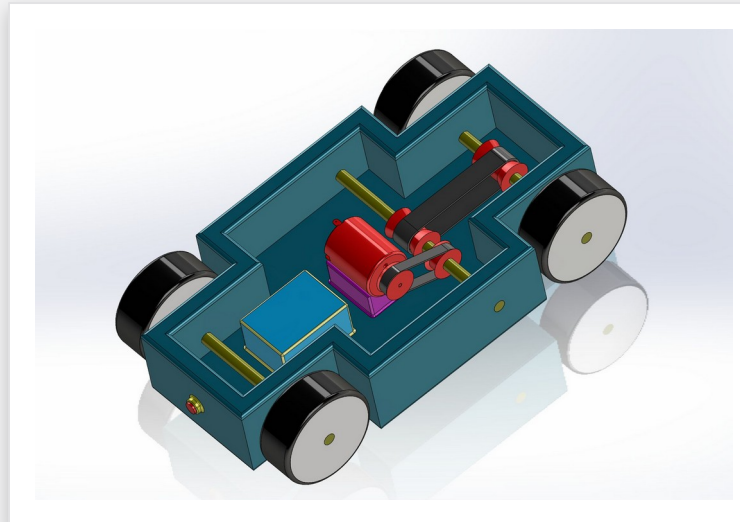
05

OVERVIEW

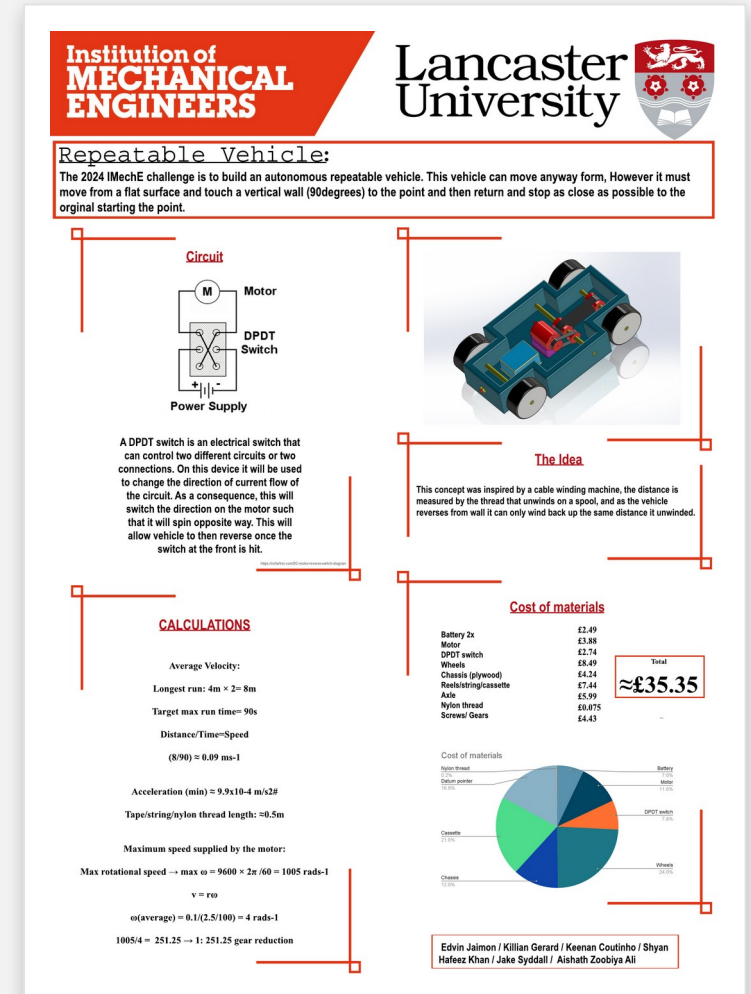
An autonomous vehicle built for the 2024 IMechE Foundation Design Challenge, capable of travelling to a vertical wall and returning to its starting point. Motor direction reversal is achieved through a DPDT switch-based circuit triggered on wall contact.

MY CONTRIBUTION

Calculated gear reduction ratios and motor specifications to meet target run time and repeatability requirements within a constrained £35 budget.



Vehicle CAD model



Design, circuit & calculations

Hydroponic Plant Growing System

Sustainability Design Project · 2019 – 2020

06

OVERVIEW

An indoor hydroponic plant growing system conceived to promote sustainable urban gardening. The design was informed by research into urban agriculture techniques and analysis of user needs and environmental constraints.

MY CONTRIBUTION

Produced multiple design iterations through sketching and CAD, then prototyped across various materials and refined with peer feedback to ensure functionality, durability and aesthetic quality.



Prototype growing system